Solving Simultaneous Equations

Exam code: 0607

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Written by: Mark Curtis
Reviewed by: Dan Finlay
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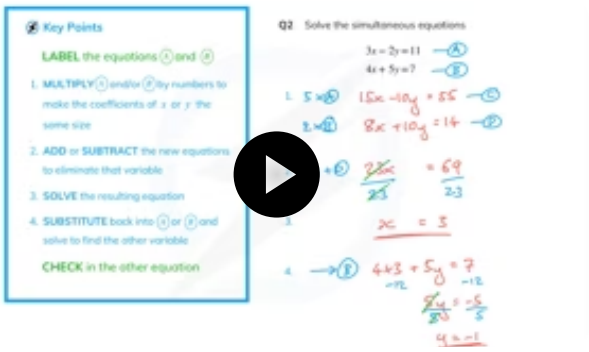

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Linear simultaneous equations

What are linear simultaneous equations?

- When there are **two unknowns** (***x* and *y***), we need **two equations** to find them both
  - For example, $3x + 2y = 11$ and $2x - y = 5$
    - The values that work for both equations at the same time are $x = 3$ and $y = 1$
- These are called **linear simultaneous equations**
  - **Linear** because there are no complicated terms like x^2 or y^2
  - **Simultaneous** because the solutions satisfy both equations at the **same time**

How do I solve linear simultaneous equations by elimination?

- **Elimination** removes one of the variables, x or y
- To eliminate the x's from $3x + 2y = 11$ and $2x - y = 5$, make the number in front of the x (the **coefficient**) in both equations the same (the sign may be different)
  - Multiply **every term** in the **first** equation **by 2**

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- **Subtracting** the second equation from the first **eliminates *x***
  - When the **sign** in front of the term you want to eliminate is the **same**, **subtract** the equations

$$\begin{array}{r} 6x + 4y = 22 \\ -\ \underline{6x - 3y = 15} \\ 7y = 7 \end{array}$$

- The y terms have become $4y - (-3y) = 7y$ (be careful with **negatives**)
  - **Solve** the resulting equation to find y
  - $y = 1$

- Then **substitute** $y = 1$ into one of the **original** equations to find x
  - $3x + 2 = 11$, so $3x = 9$, giving $x = 3$

- Write out **both solutions** together, $x = 3$ and $y = 1$
- **Alternatively**, you could have eliminated the y's from $3x + 2y = 11$ and
  $2x - y = 5$ by making the **coefficient** of y in both equations the same
  - Multiply every term in the second equation by 2
  - **Adding** this to the first equation **eliminates *y*** (and so on)
    - When the **sign** in front of the term you want to eliminate is **different**, **add** the equations

$$\begin{array}{r} 3x + 2y = 11 \\ +\ \underline{4x - 2y = 10} \\ 7x \qquad = 21 \end{array}$$

How do I solve linear simultaneous equations by substitution?

- **Substitution** means substituting one equation into the other
  - This is an **alternative** method to **elimination**
    - You can still use elimination if you prefer

- To solve $3x + 2y = 11$ and $2x - y = 5$ by substitution
  - **Rearrange** one of the equations into ***y* =** ... (or x = ...)
    - For example, the second equation becomes $y = 2x - 5$

  - **Substitute** this into the first equation
    - This means **replace** all y's with $2x - 5$ in brackets
    - $3x + 2(2x - 5) = 11$

  - **Solve** this equation to find x
    - $x = 3$

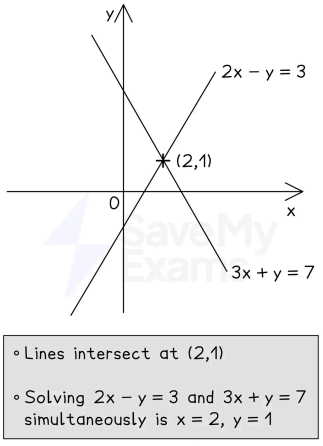
  - Then **substitute** $x = 3$ into $y = 2x - 5$ to find y
    - $y = 1$

How do I solve linear simultaneous equations graphically?

- **Plot** both equations on the same set of axes
  - To do this, you can use a **table of values**

are the x and y **coordinates** of the point of **intersection**

- For example, to solve $2x - y = 3$ and $3x + y = 4$ simultaneously
  - First **plot** them both on the **same axes** (see graph)
  - Find the **point of intersection**, (2, 1)
  - The solution is $x = 2$ and $y = 1$



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Examiner Tips and Tricks

- Always check that your final solutions satisfy **both** original simultaneous equations!
- Write out both solutions (x and y) together at the end to avoid examiners missing a solution in your working.

Worked Example

Solve the simultaneous equations

$$5x + 2y = 11$$
$$4x - 3y = 18$$

Answer:

It helps to number the equations

$$5x + 2y = 11 \quad (1)$$
$$4x - 3y = 18 \quad (2)$$

We will choose to eliminate the y terms
Make the y terms equal by multiplying all parts of equation 1 by 3 and all parts of equation 2 by 2

$$15x + 6y = 33 \quad (3)$$
$$8x - 6y = 36 \quad (4)$$

$$
\begin{array}{rl}
 & 15x + 6y = 33 \\
+ & 8x - 6y = 36 \\
\hline
 & 23x \quad = 69
\end{array}
$$

Solve the equation to find x (divide both sides by 23)

$$x = \frac{69}{23} = 3$$

Substitute $x = 3$ into either of the two original equations

$$(1) \quad 5(3) + 2y = 11$$

Solve this equation to find y

$$
\begin{aligned}
15 + 2y &= 11 \\
2y &= 11 - 15 \\
2y &= -4 \\
y &= \frac{-4}{2} = -2
\end{aligned}
$$

Substitute $x = 3$ and $y = -2$ into the other equation to check that they are correct

$$
\begin{aligned}
(2) \quad 4x - 3y &= 18 \\
4(3) - 3(-2) &= 18 \\
12 - (-6) &= 18 \\
18 &= 18
\end{aligned}
$$

Write out both solutions together

$$x = 3, \quad y = -2$$

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